CPRG 306 - Web Development 2

Description
This course builds on the fundamentals of full stack web development and includes topics such as web application architecture, front-end frameworks and back-end technologies. Students will gain hands-on experience combining front-end and back-end technologies as they develop and deploy a full stack web application.

3 Credits

Time Guidelines
The standard instructional time for this course is 75 hours.

Prerequisites

- CPRG 213

Course Assessment

Assignment	50%
Project	50%
Total	100%

Use of Turnitin

The instructor may submit student work in this course to Turnitin's text-matching software program, to help assess the academic integrity of student work in this course. Turnitin results may be considered as one piece of evidence in academic misconduct hearings. Turnitin is an American company that keeps no unencrypted student identity data in the United States unless the students themselves choose to share this information in their submission. Students should limit their sharing of personally identifiable information by not including their names and student ID numbers within the text body of submitted assignments. SAIT has carefully reviewed this company's data management procedures.

Course Learning Outcomes

1. Explain web application architecture.

    Objectives:

    1.1 Describe key concepts in web application architecture.

    1.2 Discuss the client-server relationship.

    1.3 Compare front-end and back-end frameworks.

1.4 Compare front-end and back-end technologies.

2. Apply front-end frameworks.

Objectives:

2.1 Apply popular enterprise frameworks for responsive web applications.

2.2 Apply advanced concepts in scripting.

2.3 Apply framework components.

2.4 Combine front-end technologies to maximize productivity.

3. Apply back-end technologies.

Objectives:

3.1 Describe back-end architecture.

3.2 Explore server side frameworks.

3.3 Demonstrate the purpose and functionality of server modules.

3.4 Describe REST

3.5 Use a server to connect to and manage a database.

3.6 Apply security methods to protect the web application.

4. Develop a full stack web application.

Objectives:

4.1 Analyze the business logic for a provided scenario.

4.2 Design the web application architecture for a provided scenario.

4.3 Create a database to meet system requirements.

4.4 Integrate back-end technologies with front-end technologies.

4.5 Test the functionalities of a full stack web application.

4.6 Modify web application based on testing results.

SAIT Policies and Procedures:
For information on the SAIT Grading Scale, please visit policy AC 3.1.1 Grading Progression Procedure, found on the SAIT Academic Policies and Procedures page: https://www.sait.ca/about-sait/administration/policies-and-procedures
For information on SAIT Academic Policies, please visit: www.sait.ca/about-sait/administration/policies-and-procedures/academic-student

Copyright 2015 - 2024, Southern Alberta Institute of Technology (SAIT). All Rights Reserved. This document and materials herein are protected by applicable intellectual property laws. Unauthorized reproduction and distribution of this publication in whole or part is prohibited.